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CCNA - Network Troubleshooting

- **Weekly Practice and TroubleshootingPractical (2 hours):Resolve connectivity issues in a complex network.Task: Submit troubleshooting steps and resolutions.**

  - **Assumed Network Topology**

Device/Network	IP Address / Network
PC1	192.168.10.10/24
PC2	192.168.20.10/24
VLAN 10 Gateway	192.168.10.1
VLAN 20 Gateway	192.168.20.1
Router R1	192.168.10.1, 10.0.0.1
Router R2	10.0.0.2, 192.168.20.1
Web Server	192.168.30.10
External/Internet Server	203.0.113.10

1. Use the ping command to test connectivity between two PCs on different subnets in your Packet Tracer network and document the output for both successful and failed attempts.

ANS.

- **Objective**
  To test connectivity between two PCs located on different IP subnets and document both successful and failed connectivity attempts.

- **Network Configuration**

  - **PC1**
    IP Address: 192.168.10.10
    Subnet Mask: 255.255.255.0
    Default Gateway: 192.168.10.1

  - **PC2**
    IP Address: 192.168.20.10
    Subnet Mask: 255.255.255.0
    Default Gateway: 192.168.20.1
    Because PC1 and PC2 belong to different subnets, a router or Layer-3 device is required for communication.

- **Successful Test**
  From PC1:
  ping 192.168.20.10

- **Output**
  Pinging 192.168.20.10 with 32 bytes of data:

  Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
  Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
  Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
  Reply from 192.168.20.10: bytes=32 time<1ms TTL=126

  Ping statistics for 192.168.20.10:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

- **Result**
  The ping was successful. PC1 was able to communicate with PC2 through the Layer-3 routing device.

- **Failed Test**
  To simulate a failure, the default gateway of PC1 was temporarily changed to:
  192.168.10.254
  Then:
  ping 192.168.20.10

- **Output**
  Request timed out.
  Request timed out.
  Request timed out.
  Request timed out.

  Ping statistics for 192.168.20.10:
  Packets: Sent = 4, Received = 0, Lost = 4 (100% loss

- **Diagnosis**
  The incorrect default gateway prevented PC1 from forwarding traffic to the router responsible for reaching the 192.168.20.0/24 network.

- **Corrective Action**
  The gateway was changed back to:
  192.168.10.1
  The ping was then tested again and returned successful replies.

2. Identify and fix an incorrect default gateway configuration on one device in your

network so that it can access an external web server simulation; submit the steps you took and the command outputs before and after the fix.

ANS.

- **Objective**
  To identify and correct an incorrect default gateway so that a PC can access an external web server.

- **Problem**
  PC1 was configured as:
  IP Address: 192.168.10.10
  Subnet Mask: 255.255.255.0
  Default Gateway: 192.168.10.254
  The actual router interface was:
  192.168.10.1
  Therefore, the PC could communicate only with devices on its local subnet but could not reach external networks.

- **Step 1: Check the PC Configuration**
  In Packet Tracer PC:

  - **Desktop → IP Configuration**
    Incorrect configuration:
    IP Address: 192.168.10.10
    Subnet Mask: 255.255.255.0
    Default Gateway: 192.168.10.254

- **Step 2: Test the Gateway**
  From the PC command prompt:
  ping 192.168.10.1

  - **Output**
    Request timed out.
    Request timed out.
    Request timed out.
    Request timed out.

    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
    This indicates that the configured gateway was not functioning as the PC's valid gateway.

- **Step 3: Correct the Default Gateway**

Change:
192.168.10.254
to:
192.168.10.1

- **Step 4: Test the Gateway Again**
  ping 192.168.10.1

  - **Output**
    Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
    Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
    Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
    Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

- **Step 5: Test the External Web Server**
  ping 203.0.113.10

  - **Output**
    Reply from 203.0.113.10: bytes=32 time=20ms TTL=120
    Reply from 203.0.113.10: bytes=32 time=19ms TTL=120
    Reply from 203.0.113.10: bytes=32 time=21ms TTL=120
    Reply from 203.0.113.10: bytes=32 time=20ms TTL=120

    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Test	Before Fix	After Fix
Ping Gateway	Failed	Successful
Ping External Server	Failed	Successful
Packet Loss	100%	0%
Default Gateway	192.168.10.254	192.168.10.1

3. A user reports they cannot access Flipkart.com from their simulated network. Trace the path using traceroute (or tracert) and identify where the connection fails; explain how

you diagnosed and resolved the issue.

ANS.

- **Objective**
  To trace the network path and determine where connectivity to the simulated Flipkart server fails. In Packet Tracer, an actual public website such as Flipkart cannot normally be traced over the real Internet. Therefore, this task is demonstrated using a **simulated Flipkart web server** in the Packet Tracer topology.

  Assume:
  Simulated Flipkart Server: 203.0.113.20

- **Step 1: Test Connectivity**
  From the PC:
  ping 203.0.113.20

  - **Output**
    Request timed out.
    Request timed out.
    Request timed out.
    Request timed out.

    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

- **Step 2: Trace the Path**
  Use:
  tracert 203.0.113.20

  - **Sample Output**
    Tracing route to 203.0.113.20

    1 <1 ms 192.168.10.1
    2 5 ms 10.0.0.2
    3 * Request timed out.
    4 * Request timed out.

    Trace complete.

  - **Diagnosis**
    The first two hops respond:
    PC
    ↓

192.168.10.1
↓
10.0.0.2
↓
X Failure
This indicates that the PC and the first routers are functioning, but the route beyond R2 is unavailable.

- **Step 3: Check the Router Routing Table**
  On R2:
  show ip route

  - **Incorrect Output**
    C 10.0.0.0/30 is directly connected
    C 192.168.20.0/24 is directly connected
    There is no route to:
    203.0.113.0/24
    Therefore, R2 does not know where to forward packets destined for the simulated Flipkart server.

- **Step 4: Add the Correct Route**
  For example:
  R2(config)# ip route 203.0.113.0 255.255.255.0 10.0.0.1
  Then verify:
  R2# show ip route
  The routing table should now contain a route similar to:
  S 203.0.113.0/24 [1/0] via 10.0.0.1

- **Step 5: Test Again**
  ping 203.0.113.20

  - **Output**
    Reply from 203.0.113.20: bytes=32 time=20ms TTL=118
    Reply from 203.0.113.20: bytes=32 time=19ms TTL=118
    Reply from 203.0.113.20: bytes=32 time=21ms TTL=118
    Reply from 203.0.113.20: bytes=32 time=20ms TTL=118

    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

4. Simulate a scenario where a router has an incorrect static route that breaks connectivity between two VLANs. Correct the route configuration and verify the solution using appropriate show commands. Hint: Use 'show ip route' and 'show

running-config' to check current routes and settings.

ANS.

- **Objective**
  To identify and correct an incorrect static route that prevents communication between two VLANs.

  - **Network**
    VLAN 10 → 192.168.10.0/24
    VLAN 20 → 192.168.20.0/24
    Assume Router R1 has an incorrect route:
    ip route 192.168.20.0 255.255.255.0 10.0.0.5
    But the correct next-hop router is:
    10.0.0.2

- **Step 1: Check the Routing Table**
  On R1:
  R1# show ip route

  - **Incorrect Output**
    C 192.168.10.0/24 is directly connected
    C 10.0.0.0/30 is directly connected
    S 192.168.20.0/24 [1/0] via 10.0.0.5
    The next-hop address 10.0.0.5 is incorrect.

- **Step 2: Check Running Configuration**
  R1# show running-config
  The incorrect configuration is found:
  ip route 192.168.20.0 255.255.255.0 10.0.0.5

- **Step 3: Remove the Incorrect Route**
  R1(config)# no ip route 192.168.20.0 255.255.255.0 10.0.0.5

- **Step 4: Configure the Correct Route**
  R1(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2

- **Step 5: Verify the Route**
  R1# show ip route

  - **Correct Output**
    C 192.168.10.0/24 is directly connected
    C 10.0.0.0/30 is directly connected
    S 192.168.20.0/24 [1/0] via 10.0.0.2
    The S indicates a **static route**.

- **Step 6: Test Connectivity**
  From PC1:
  ping 192.168.20.10

  - **Output**
    Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
    Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
    Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
    Reply from 192.168.20.10: bytes=32 time<1ms TTL=126

    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

5. You receive a network diagram and a set of running-config outputs for a multi-router setup where WhatsApp voice calls are not working between two branches. Analyze the configs, identify the misconfiguration, and describe your troubleshooting steps and resolution.

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- **Scenario**
  A company has two branches:
  Branch A Branch B
  192.168.10.0/24 192.168.20.0/24
  | |
  R1 ============================== R2
  WAN Link
  Users can successfully ping each other, but **WhatsApp voice calls are not working** between the two branches.
  The router configurations are analyzed to find the cause.

- **Step 1: Check Basic Connectivity**
  From Branch A:
  ping 192.168.20.10

  - **Result**
    Reply from 192.168.20.10
    Reply from 192.168.20.10
    Reply from 192.168.20.10
    Reply from 192.168.20.10

    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
    This proves that basic IP connectivity exists.

- **Step 2: Check Routing Tables**
  On R1:
  show ip route

On R2:
show ip route
Both routers contain routes to the remote branch.
Therefore, the problem is probably not basic routing.

- **Step 3: Check Running Configuration**
  Use:
  show running-config
  During configuration analysis, the following ACL entry is found:
  access-list 101 deny udp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
  This rule blocks UDP traffic between the two branch networks.
  Because voice communication commonly uses UDP-based media traffic, this ACL can prevent the simulated voice traffic even though ICMP ping works.

- **Step 4: Identify the Misconfiguration**
  The problematic configuration is:
  access-list 101 deny udp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
  The ACL is blocking UDP traffic from Branch A to Branch B.

- **Step 5: Correct the ACL**
  Remove the incorrect rule:
  R1(config)# no access-list 101 deny udp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
  Then ensure that legitimate traffic is permitted according to the organization's intended security policy.
  For a simple Packet Tracer demonstration, an appropriate permit statement could be:
  R1(config)# access-list 101 permit udp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
  If ACL 101 is applied to an interface, verify its placement:
  show running-config
  Look for:
  ip access-group 101 in
  or:
  ip access-group 101 out

- **Step 6: Verify the ACL**
  R1# show access-lists

  - **Example**
    Extended IP access list 101
      permit udp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255

- **Step 7: Verify Connectivity Again**
  Test the branch-to-branch connection:
  ping 192.168.20.10
  Then use Packet Tracer's simulation tools or the relevant application/service simulation to verify that the intended UDP traffic is now permitted.

Task	Problem	Diagnostic Command	Solution	Verification

1	PC-to-PC connectivity failure	ping	Correct gateway/routing	Successful ping
2	Incorrect default gateway	ping gateway	Correct gateway address	External server ping
3	Simulated Flipkart unreachable	tracert, show ip route	Add missing route	Successful ping/trace
4	Incorrect static route	show ip route, show running-config	Replace next-hop address	Cross-VLAN ping
5	Voice traffic blocked	show access-lists, show running-config	Correct ACL	UDP traffic permitted